Microsoft Windows workloads. Red Hat Enterprise Virtualization infrastructures are based on KVM hypervisor (kernel-based virtual machine) and features management tools that virtualise resources, processes, and applications to ensure service performance and business continuity. RHV uses the SPICE protocol and VDSM (virtual desktop server manager) with an RHEL-based centralised management server. The OpenShift container platform is its successor.

Workplace deployment automation and administration

Enterprises need Infrastructure as Code (IaC) tools to automate workplace deployment and provisioning. There are open source tools available for this purpose too. These tools have evolved and matured to meet the diversified demands of enterprises.

- ***Terraform:*** IaC removes the human component of setting up servers, virtual machines, and provisioned cloud computing through an automated process. Terraform is a free and open source tool created by HashiCorp. It is also one of the most popular IaC tools. Terraform makes a pre-execution check to ensure the desired result has been achieved. It is sometimes used as a base tool in conjunction with Ansible. It is highly compatible with cloud services such as AWS, Google Cloud, and Microsoft Azure.
- ***OpenStack:*** OpenStack is a free, open standard cloud computing platform. It is mostly deployed as Infrastructure-as-a-Service (IaaS) in both public and private clouds, where virtual servers and other resources are made available to users. The software platform consists of interrelated components that control diverse, multi-vendor hardware pools of processing, storage, and networking resources throughout a data center. Users manage it either through a web-based dashboard, through command-line tools, or through RESTful web services. Beyond standard Infrastructure-as-a-Service functionality, additional components provide orchestration, fault management and service management amongst other services to ensure high availability of user applications.

Puppet and Chef are two other popular open source tools used for workplace deployment and provisioning.

Workplace support tools

Workplace support teams need many tools to support end users. Three open source tools that can be chosen as alternatives to commercial tools for this purpose are listed below.

Open source helpdesk is software that offers a ticket management system. Built on open source technology, it is a great solution for offering efficient customer support.

OSTicket is an open source helpdesk and ticketing system used by more than 5 million users worldwide, and by more than 15,000 businesses.

Rocket.chat is an open source versatile communication platform that enables you to engage in seamless interactions regardless of how end users connect.

DevOps workplaces

Most DevOps workplaces make use of containers as a primary resource. These workplaces require tools covering container builds, orchestration, microservices networking, configuration management, CI/CD automation, full-stack monitoring and more. Popular open source tools deployed in DevOps workplaces are:

Kubernetes: It is used to orchestrate containers. Instead of releasing microservices manually, Kubernetes can automate deployment, maintenance and scaling of groups of containers in production. It is hosted by the Cloud-Native Computing Foundation (CNCF).

Docker: This free and open source platform is used to build, ship and run an application as a lightweight container. Containers package up the binaries, libraries, configuration files and dependencies required for a program to run. It makes configuration management, issue control, and scaling much easier through the use of containers that can be moved from place to place.

Istio: Istio enables organisations to secure, connect, and monitor microservices, so that they can modernise their enterprise apps swiftly and securely. It manages traffic flows between services, enforces access policies, and aggregates telemetry data, all without requiring changes to application code.

Ansible, Puppet, Jenkins, and Git are a few other popular open source tools used in DevOps workplaces. END

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